

Perimeter, Surface Area, Volume Questions

(1)

Example 1: A farmer wants to build a small pen for his chickens. One side of the pen will be against his barn. If he has 80 metres of fencing to use, what is the maximum possible area?

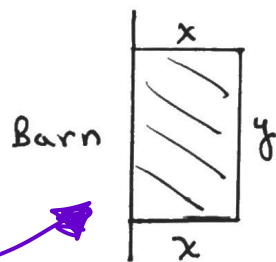
$L = 40$
 $W = 20$



Solution:

Let x be the width of the pen.

Let y be the length of the pen.



$$P = 2x + y$$

$$80 = 2x + y$$

$$y = 80 - 2x$$

$A(x)$

$$P(x) = xy$$

$$= x(80 - 2x)$$

$$P'(x) = 80 - 4x$$

$$P'(x) = 0$$

$$80 - 4x = 0$$

$$\therefore x = 20 \Rightarrow y = 40$$

optional

$$* \text{ check: } P''(20) = -4 < 0 \Rightarrow \text{max.}$$

$$\therefore \text{Maximum area is } 20\text{m} \times 40\text{m} = 800\text{m}^2$$

m^2 or units^2

(2)

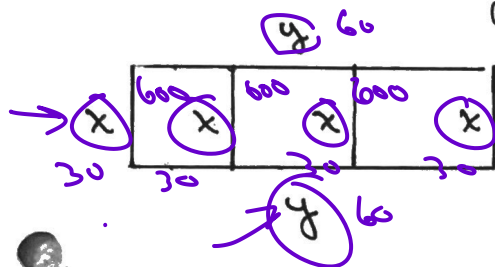
Example 2: A rancher wants to enclose 3 equal adjacent rectangular lots with an area of 600 m^2 each. What is the least amount of fencing required?

→ perimeter

Solution:

Let x be the width of each lot.

Let y be the total length of the lots.



$$A = x \cdot y$$

$$1800 = x \cdot y$$

$$y = \frac{1800}{x}$$

$$P(x) = 4x + 2y$$

$$P(x) = 4x + \frac{3600}{x}$$

$$P'(x) = 4 - \frac{3600}{x^2}$$

$$P'(x) = 0$$

$$4 - \frac{3600}{x^2} = 0$$

$$x^2 = 900$$

$$x = \pm 30 \quad (-30 \text{ is inadmissible})$$

$$y = 60$$

* check: $P''(30) = \frac{7200}{(30)^3}$

$$= 0.27 > 0 \Rightarrow \text{min.}$$

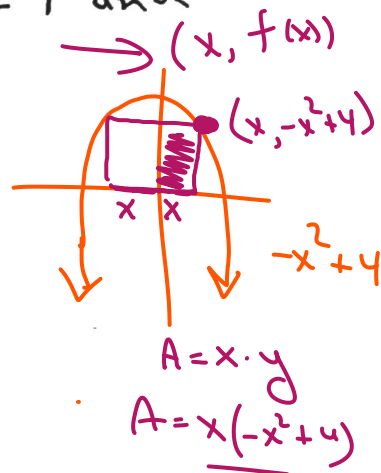
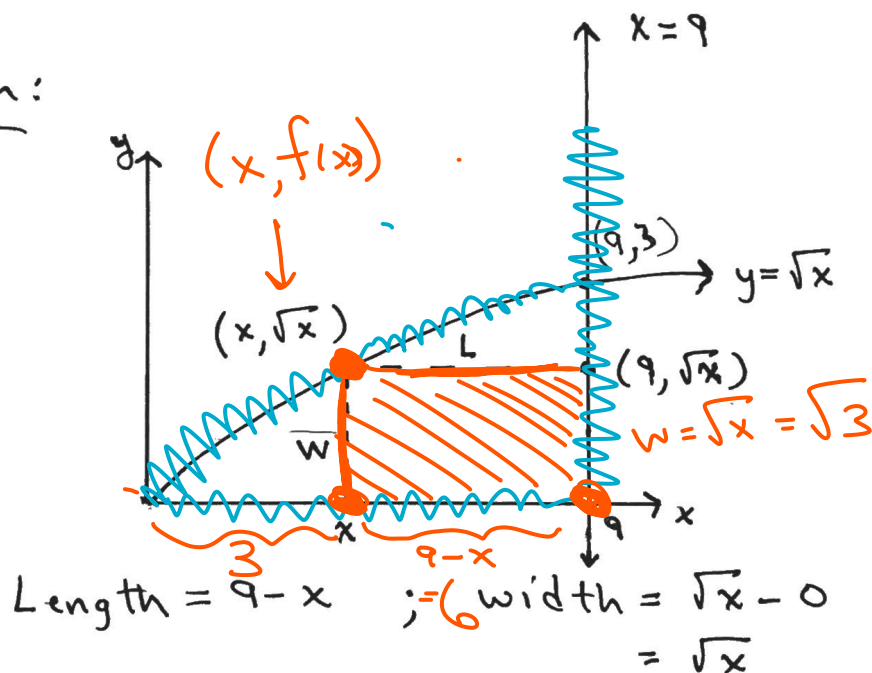
$\therefore$ The amount of fencing is $4(30) + 2(60) = 240 \text{ m}$

optional

(3)

Example 3: Determine the dimensions of the rectangle of largest area which can be inscribed in the closed region bounded by the x -axis, $x=9$ and $y=\sqrt{x}$.

Solution:



$$A = L \cdot w$$

$$A = (9 - x)\sqrt{x}$$

$$A' = -\sqrt{x} + \frac{9 - x}{2\sqrt{x}}$$

$$A' = 0$$

$$-\sqrt{x} + \frac{9 - x}{2\sqrt{x}} = 0$$

$$-2x + 9 - x = 0$$

$$3x = 9$$

$$x = 3$$

$$\therefore w = \sqrt{x} ; L = 9 - x$$

$$w = \sqrt{3} \quad L = 9 - 3$$

$$L = 6$$

$$\therefore \text{Area} = 6 \times \sqrt{3} \text{ units}^2$$

Example 4: A manufacturer wishes to produce cylindrical fruit cans with a capacity of ~~250 mL~~ [✓]. What dimensions will minimize the amount of material required for the can?

Solution:

$$V = 250 \text{ mL}$$

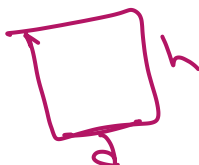
$$250 = \pi r^2 h$$

$$h = \frac{250}{\pi r^2}$$

h in terms of r

$$h = 2r$$

$$h = \text{diameter}$$



$$\therefore r \doteq 3.41 \text{ cm and}$$

$$h \doteq 6.83 \text{ cm.}$$

$$S.A. = 2\pi r^2 + 2\pi r h$$

top + bottom side

$$S.A. = 2\pi r^2 + 2\pi r \left(\frac{250}{\pi r^2} \right)$$

$$S.A. = 2\pi r^2 + \frac{500}{r}$$

$$S.A.' = 4\pi r - \frac{500}{r^2}$$

$$S.A.' = 0$$

$$4\pi r - \frac{500}{r^2} = 0$$

$$4\pi r^3 - 500 = 0$$

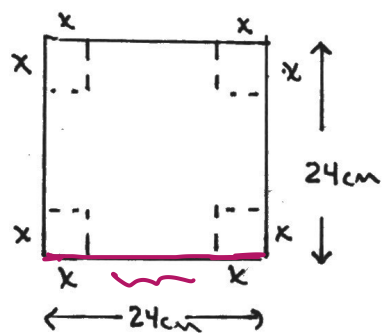
$$r^3 = \frac{500}{4\pi}$$

$$r = \sqrt[3]{\frac{500}{4\pi}} \doteq \underline{3.41 \text{ cm}}$$

$$\Rightarrow h \doteq \underline{6.83 \text{ cm}}$$

Example 5: An open topped box is to be made from a square piece of cardboard that is 24 cm x 24 cm by cutting equal squares from each corner and folding up the flaps to make sides. What is the greatest volume for the box?

Solution:



Let $x = \text{height}$

Let $24 - 2x = \text{length} = \text{width}$

$$V = L \cdot w \cdot h$$

$$V = (24 - 2x)^2 \cdot x$$

$$V' = 2(24 - 2x)(-2)(x) + (24 - 2x)^2$$

$$V' = (24 - 2x)[-4x + (24 - 2x)]$$

$$V' = (24 - 2x)(-6x + 24)$$

$$V' = -12(12 - x)(x - 4)$$

$$V' = 0$$

$$-12(12 - x)(x - 4) = 0$$

$$\therefore x = 12, x = 4$$

↑
inadmissible.

$$\therefore h = 4 \text{ cm}$$

$$L = W = 16 \text{ cm}$$

$$\text{Volume} = 16 \text{ cm} \times 16 \text{ cm} \times 4 \text{ cm}$$

$$\text{Volume} = 1024 \text{ cm}^3$$

Restrictions

$$0 < x < 12$$

Example 6: Determine the dimensions of the cylinder of maximum volume that can be inscribed in a sphere of radius R cm.

Let R = radius of the sphere.

note: $h = 2x$

$$x = \pm \frac{10}{\sqrt{3}} \quad \left(-\frac{10}{\sqrt{3}} \text{ is inadmissible} \right)$$

and

$$r = \sqrt{100 - \frac{100}{3}} = \sqrt{\frac{200}{3}} \text{ cm}$$